Integration

Section 1: Standard Integrals (Trig, Exp & Rational)

1 Find $\int \frac{2}{3x-1}\,dx$.

a. $\frac{2}{3}\ln|3x-1|+C$

b. $2\ln|3x-1|+C$

c. $\frac{1}{3}\ln|3x-1|+C$

d. $-\frac{2}{(3x-1)^2}+C$

2 Find $\int \cos^2(x)\,dx$.

a. $\frac{x}{2}+\frac{\sin(2x)}{4}+C$

b. $\frac{\sin^3(x)}{3}+C$

c. $\frac{x}{2}-\frac{\sin(2x)}{4}+C$

d. $\sin(2x)+C$

3 Evaluate $\int_0^{\pi/4} \tan(x)\,dx$.

a. $\frac{1}{2}\ln 2$

b. $\ln 2$

c. 1

d. $\frac{\sqrt{2}}{2}$

Section 2: Integration by Substitution

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4 Evaluate $\int x\sqrt{x^2+5}\,dx$.

a. $\frac{1}{3}(x^2+5)^{3/2}+C$

b. $\frac{1}{2}(x^2+5)^{3/2}+C$

c. $\frac{2}{3}(x^2+5)^{3/2}+C$

d. $(x^2+5)^{3/2}+C$

5 Evaluate $\int_0^1 \frac{x}{x^2+1}\,dx$.

a. $\frac{1}{2}\ln 2$

b. $\ln 2$

c. $\frac{\pi}{4}$

d. 1

6 Use the substitution $x = 2\sin\theta$ to find $\int \sqrt{4-x^2}\,dx$.

a. $2\theta + \sin(2\theta) + C$

b. $4\theta + 2\sin(2\theta) + C$

c. $2\arcsin(x/2) + C$

d. $\frac{1}{2}(4-x^2)^{3/2}+C$

Section 3: Integration by Parts

7 Find $\int xe^{3x}\,dx$.

a. $\frac{xe^{3x}}{3} - \frac{e^{3x}}{9} + C$

b. $\frac{xe^{3x}}{3} - \frac{e^{3x}}{3} + C$

c. $xe^{3x} - e^{3x} + C$

d. $\frac{x^2e^{3x}}{6} + C$

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8 Evaluate $\int_1^e x\ln x\,dx$.

a. $\frac{e^2+1}{4}$

b. $\frac{e^2-1}{4}$

c. $\frac{e^2}{2}$

d. $\frac{e^2-1}{2}$

9 Evaluate $\int_0^\pi x\sin x\,dx$.

a. π

b. 2π

c. 0

d. 2

Section 4: Partial Fractions Integration & Trapezium Rule

10 Find $\int \frac{5x-2}{(x-2)(x+1)}\,dx$.

a. $\frac{8}{3}\ln|x-2|+\frac{7}{3}\ln|x+1|+C$

b. $3\ln|x-2|+2\ln|x+1|+C$

c. $8\ln|x-2|+7\ln|x+1|+C$

d. $\ln|(x-2)(x+1)|+C$

11 Use the trapezium rule with 4 intervals (5 ordinates) to approximate $\int_1^3 \frac{1}{x}\,dx$.

a. 1.117

b. 1.099

c. 1.150

d. 1.050

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12 Is the trapezium rule approximation in the previous question an overestimate or underestimate, and why?

a. Overestimate, because the curve $y = 1/x$ is concave up

b. Underestimate, because $y = 1/x$ is decreasing

c. Overestimate, because the intervals are too wide

d. Underestimate, because the chords lie below the curve

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